

ORIGINAL ARTICLE

Uric acid as a risk factor in the progression of arterial hypertension

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ABSTRACT

Background and aim: The impact of elevated serum uric acid (UA) levels on the course of hypertension and the development of cardiovascular complications remains controversial in international studies. This study aimed to evaluate the association between serum UA levels and clinical parameters of patients with arterial hypertension (AH) and to assess the influence of age and clinical characteristics on UA levels.

Methods: A single-stage retrospective cross-sectional analysis was conducted, including 187 patients (out of 1,112 medical records) undergoing follow-up for hypertension at municipal polyclinics Nos. 1, 6, 7, and 10 in Astana, with records comprising age, sex, anthropometric measurements, and clinical characteristics, including biochemical blood tests for UA. Descriptive statistics were performed to summarize and analyze the results qualitatively.

Results: The study results indicated that the maximum permissible UA level was 360 $\mu\text{mol/L}$ in women and 420 $\mu\text{mol/L}$ in men. Among the participants, 9 women (7.8%) exhibited hyperuricemia (HU), with a maximum UA value of 387 $\mu\text{mol/L}$. No significant effect of gender or age on UA levels was observed ($p>0.05$). Regression analysis showed that in patients with elevated systolic blood pressure (SBP >140 mmHg), a 1 $\mu\text{mol/L}$ increase in UA was associated with a 0.02 mmHg increase in SBP ($p=0.04$). In a subgroup analysis, an increase in body mass index (BMI) by 1 kg/m^2 was associated with a 1.83 $\mu\text{mol/L}$ increase in UA levels ($p=0.006$).

Conclusions: Overall, the proportion of patients with elevated UA was low. Additionally, UA levels gradually increased in patients aged >50 years (≥ 296 $\mu\text{mol/L}$), potentially reflecting age-related hyperproduction of UA. These findings highlight the descriptive trends in UA distribution across demographic and clinical subgroups, while acknowledging the relatively low prevalence of clinically significant hyperuricemia in this cohort. (www.actabiomedica.it)

Key words: prooxidant effect, kidney disease, overweight, body mass index, atherosclerosis



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Introduction

Uric acid (UA) has a notable antioxidant capacity due to its ability to chelate transition metals. Simultaneously, it is generated by the enzyme xanthine oxidoreductase, which converts hypoxanthine to xanthine and subsequently to urate radicals, which can induce oxidative reactions via a prooxidant effect (1). High UA levels are believed to provoke endothelial dysfunction in renal vessels, contributing to the development and persistence of elevated blood pressure (BP) (2,3). This mechanism is frequently cited as a key explanation for the association between elevated serum UA and cardiovascular risk factors (4). Secondary hyperuricemia arises from imbalances in UA production and excretion, due to increased cellular turnover in metabolic disorders or impaired elimination in chronic kidney disease. Exogenous factors, including alcohol, red meat, seafood, and other purine-rich foods, can further elevate UA, while endogenous synthesis occurs in the liver, intestines, muscles, and vascular endothelium, providing a rationale for observed links between hyperuricemia (HU) and risk factors such as overweight, smoking, and dietary habits (5).

Conversely, other studies have reported that low UA may be associated with increased mortality among hemodialysis patients (6,7). The relationship between UA and cardiovascular disease is noted not only in severe hypertension (defined as >6 mg/dL [360 μ mol/L]) in women and >7 mg/dL [420 μ mol/L] in men) but also at levels considered high-normal (>5.2 to 5.5 mg/dL [310 – 330 μ mol/L]) (8,9). Despite experimental evidence demonstrating that HU can provoke endothelial dysfunction by promoting oxidative stress and inflammation, the exact role of UA in hypertension development remains unclear, and the literature provides inconsistent and poorly characterized evidence (10).

Given these gaps, the significance of UA as a causal risk factor versus a correlated marker is still debated, highlighting the need for further clinical investigation. This study aims to clarify potential associations between UA levels and arterial hypertension (AH) in patients, and to examine the influence of clinical and demographic characteristics on these levels. Improved understanding of these relationships may

assist clinicians in assessing patients with hypertension who present with elevated UA, tailoring interventions more effectively while considering comorbidities such as obesity, atherosclerosis, and chronic kidney disease.

Materials and methods

In the study, subgroup analysis was rationalized by defining groups according to age (31–40, 41–50, 51–60, 61–70, >70 years), sex (men $n=71$, women $n=116$), body mass index (normal weight $n=15$, overweight $n=133$, obesity grade I $n=32$, obesity grade II $n=7$), and systolic blood pressure (SBP <140 mmHg $n=135$, SBP >140 mmHg $n=52$). Each subgroup was clinically justified, given the potential influence of age, sex, and body mass on serum uric acid levels and the progression of hypertension. To reduce the risk of false associations, correction for multiple testing (Bonferroni) was applied, allowing reliable assessment of statistical significance in both small and large subgroups and providing a transparent and representative evaluation of the relationship between UA and AH.

A detailed comparison was conducted between the full cohort ($n=1112$) and the analyzed sub-sample with available uric acid measurements ($n=187$) to assess representativeness: the sub-sample included 72 men and 115 women, with a mean age of 59 years, mean BMI of 28 kg/m², mean systolic blood pressure of 143 mmHg, and diastolic blood pressure of 98 mmHg, which broadly reflected the distribution of characteristics in the full cohort; however, the selective availability of uric acid measurements could introduce bias, as patients with available data may have differed in clinical monitoring, comorbidities, or access to healthcare, potentially affecting the observed associations between uric acid levels and hypertension characteristics.

A single-centre cross-sectional study was conducted by retrospectively analysing the gender, age, and clinical parameters of patients assigned to city polyclinics Nos. 1, 6, 7, and 10 of the Astana Akimat, using the Damumed electronic database. To obtain permission to access the database, requests were made to the heads of these medical organisations from the Department of Science of Astana Medical University.

The list of patients under dynamic observation was obtained from the registry office in Microsoft Excel format. A total of 1112 medical records were analysed using the hypertension filter, with ICD codes I.11, I.12, I.13.

The inclusion criteria were age over 18 years; registration for hypertension for more than 5 years; and the following data: demographic and anthropometric characteristics; clinical description with blood pressure levels; and results of biochemical blood tests for UA. It is also worth noting that the patient age range for the analysis was selected based on hypertension risk. Descriptive statistics were conducted for qualitative analysis of the results using Microsoft Excel software. To characterise the variables, the mean values with standard deviation are taken. The rationale for assigning participants to groups is described in the results for each analysis. Quantitative analysis – the statistical significance of differences was determined using Student's t-test, and regression analysis was performed at a 95% confidence level using IBM SPSS Statistics 20. Patients' comorbidities, such as atherosclerosis of blood vessels, overweight, excessive consumption of salty foods and diabetes, were also accounted for. The study employed a retrospective cross-sectional design, analysing clinical and biochemical data from patients with arterial hypertension. Due to the observational and cross-sectional nature of the analysis, causal relationships cannot be established. Instead, serum uric acid (UA) levels were evaluated as a marker associated with hypertension severity and related clinical parameters, highlighting potential correlations with systolic blood pressure, body mass index, and age, rather than implying a direct causal effect on the progression of hypertension.

The study also analysed available research and compared the results with other authors. In the source analysis, 30 literature sources were selected, including 29 published in the last 5 years and 1 from 2019, to compare the development of information sources over 20–30 years (11). After that, the data obtained were fully elaborated and compared with the data of the authors' study, where the main opinions of the authors on comorbidities, pregnancy, as well as other laboratory parameters were identified. The quantitative correlation between female and male patients was

also determined, as well as the difference in cases of elevated UA levels and the effect of the index on vascular elasticity with increased systolic and diastolic blood pressure in the human body.

Regression analyses were conducted using multiple linear regression models to evaluate the associations between serum uric acid (UA) levels and systolic blood pressure (SBP), as well as between UA and body mass index (BMI). The models included relevant covariates such as age, sex, and BMI to account for potential confounding. All assumptions of linear regression, including linearity, homoscedasticity, normality of residuals, and absence of multicollinearity, were formally assessed and met. Regression coefficients are presented with 95% confidence intervals to provide measures of precision and facilitate interpretation. Tabular coefficients are fully consistent with textual descriptions, ensuring alignment between reported values and conclusions drawn regarding the relationships of UA with SBP and BMI. Moreover, concomitant pathologies in the patients were addressed, as well as arterial hypertension, acute or chronic. The study also analysed the patient's family history, which directly impacts the heritability of diseases such as diabetes, atherosclerosis and obesity. In fact, in these diseases, the possibility of BP increase in the blood is noted. The analysis of the obtained data was compared with the data of other researchers and the main differences were identified, as well as the importance of the results obtained in clinical cases.

Results

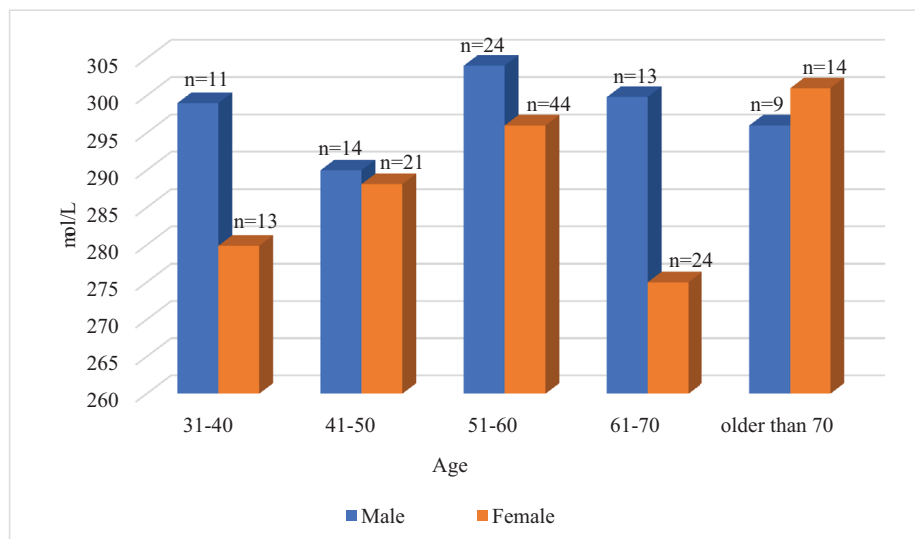
In the study, an analysis of 1112 medical records selected according to the inclusion criteria was conducted. For the study, data from 187 patients with data on serum UA levels were selected. Thus, the sample amounted to 16.8% of the total number of patients undergoing outpatient observation for hypertension. The data were grouped by risk factors for the development of hypertension – gender, age, body mass index (BMI), and blood pressure. The age and clinical characteristics of the patients are presented in Table 1.

As shown in Table 1, female participants had higher UA levels than male participants, but a

Table 1. Gender, age and clinical and laboratory characteristics of patients (mean).

Gender	Amount	Age	BMI	SBP	Diastolic blood pressure	UA level
Male	37.9% (71)	61.3±6.55	27.5±2.22	146±24.3	99.1±4.4	276.42±43.3
Female	62.1% (116)	57.5±4.1	28.7±5.54	140.33±14.9	97.6±5.1	295.76±47.2

Source: compiled by the author.

**Figure 1.** Average UA value depending on age and gender of participants.

Note: $p < 0.05$.

Source: compiled by the author.

Student's t-test showed no statistically significant difference ($p > 0.05$). Next, the participants were divided into groups based on age, BMI, and systolic blood pressure (SBP) level. The observed differences in serum uric acid (UA) levels between female and male participants can be presented as descriptive findings, without implying causality. On average, women demonstrated slightly higher UA values compared to men; however, statistical analysis indicated that these differences were not significant. These observations may be influenced by physiological factors such as menopausal status, which is associated with changes in estrogen levels affecting uric acid excretion, as well as renal function, which modulates UA clearance. Notably, the prevalence of true hyperuricemia, defined by standard thresholds ($\geq 360 \mu\text{mol/L}$ for women and $\geq 420 \mu\text{mol/L}$ for men), remained low, affecting 7.8% of the study population.

Such descriptive reporting emphasizes the limited frequency of elevated UA within the cohort while acknowledging biological and demographic factors that could account for subtle sex differences. These results highlight that, in a general hypertensive outpatient population, elevated UA is relatively uncommon, and observed variations should be interpreted cautiously, considering potential confounders. Below are the average UA values by participant age and sex to assess whether there is a uricosuric effect of estrogenic activity, according to the literature (9, 10) (Figure 1).

As shown in Figure 1, patients over 50 years of age have higher levels of UA. Next, a regression analysis of the impact of age on the level of UA was conducted (Table 2).

According to the above data, at a 95% confidence level, in all age groups, there is no statistical significance of the effect of age on the level of UA ($p > 0.05$),

which does not refute the null hypothesis that age as a risk factor for the development of HU does not influence (8). The next risk factor is BMI, which is also considered in the groups of patients with normal, overweight, first and second-degree obesity. The distribution of UA levels depending on the height and weight of patients and depending on gender is shown in Figure 2.

The results of the study showed that the maximum average UA value of 319.57 $\mu\text{mol/L}$ was found in the group of patients with normal body weight (BMI 20-25 kg/m^2) and the minimum value of 255 $\mu\text{mol/L}$ was found in patients with second-degree obesity (BMI 36-40 kg/m^2). The one-factor analysis of

variance showed the statistical significance of the differences in the indicators of these groups ($p=0.035$). Nevertheless, correlation and regression analyses were conducted to determine the trend in the level of UA depending on BMI (Table 3, 4).

According to the results of the study, BMI has an impact in patients with normal weight (direct strong correlation of 0.83) and in second-degree obesity (direct medium correlation of 0.52). Table 4 shows that, at the 95% confidence level, the level of UA significantly increases by 9.3 $\mu\text{mol/l}$ with an increase in BMI by 1 kg/m^2 ($p=0.01$) in patients with normal weight. This suggests that asymptomatic HU is a precursor to further metabolic disorders (10).

The study participants were then divided into 3 groups according to the severity of hypertension. The median values for each group and the results of the comparative analysis using the Kruskal-Wallis test are shown in Table 5.

The results of the study showed that at the confidence level of 95%, UA values do not have statistically significant differences depending on the degree of AH. At the early stage of AH diastolic BP has a greater prognostic value in younger individuals, but in elderly individuals SBP is undoubtedly the most important marker of cardiovascular complications (11). Q. Wang

Table 2. Regression coefficients of UA level depending on age group.

Age of patients	Regression coefficient	p-value
31-40	-0.06	0.088
41-50	0.015	0.26
51-60	0.001	0.81
61-70	0.007	0.835
Older than 70	0.014	0.381

Source: compiled by the author.

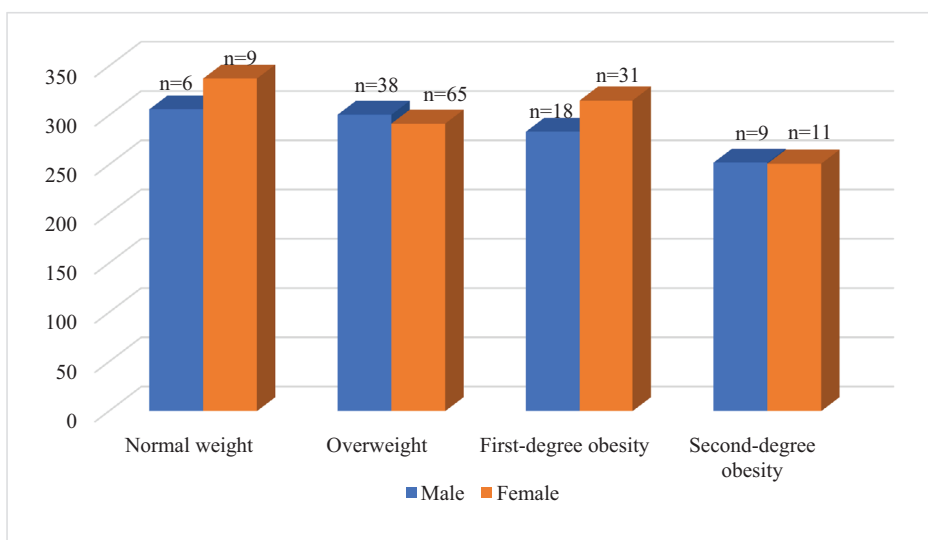


Figure 2. Average UA levels according to BMI and gender of participants.

Note: $p<0.05$.

Source: compiled by the author.

Table 3. Correlation coefficients of UA level depending on BMI.

Body Mass Index	Normal weight	Overweight	First-degree obesity	Second-degree obesity
Value	0.83	-0.1	-0.12	0.52

Source: compiled by the author.

Table 4. Regression coefficients of UA level depending on BMI.

Age	Coefficient	p-value
20-25 (8%)	9.3	0.01
26-30 (71%)	-0.4	0.46
31-35 (17%)	2.7	0.505
36-40 (3%)	0.2	0.55

Source: compiled by the author.

Table 5. Comparative analysis of clinical parameters of participants depending on the degree of hypertension.

AH degrees	Grade 1 hypertension; n=85 (45.5%)	Grade 2 hypertension; n=59 (31.5%)	Grade 3 hypertension; n=43 (30%)
Value	140-159 and/or 90-99 mmHg	160-179 and/or 100-109 mmHg	≥180 and/or ≥110 mmHg
UA (μmol/l)	296.5	293.3	303.8
p-value	>0.05	>0.05	>0.05

Source: compiled by the author.

et al. (12) HU predicted uncontrolled AH even after 3 months of treatment with fimasartan in patients with AH. For the study, patients registered for hypertension were divided into 2 groups according to their BP level at the time of the study. Patients who reached the target value of SBP<140 mmHg were 72%, and patients with stable high values of CAD>140 mmHg were 28% of the study population. The level of UA as a function of the level of SBP is shown in Figure 3.

A comparative analysis of the indicators of the two groups using the Student's t-test showed that the differences in the two groups were statistically significant. In regression analysis, the study determined that at the 95% confidence level, in the group of patients with elevated CAD (>140 mmHg), a 1 μmol/L increase in UA level leads to a 2 mmHg increase in SBP. ($p=0.04$). In patients with SBP<140mmHg, the statistical significance of the regression coefficient was

not found ($p=0.25$). The regression coefficients are presented in Table 6.

UA levels are considered as elevated at values above 360 μmol/l in women and 420 μmol/l in men (10). Following the EULAR guidelines, characteristics of patients with UA above 360 μmol/L were analysed (13). According to the results of the study, out of a sample of 187 patients with hypertension, the maximum values of UA levels were found in 12 patients (6.4%), in particular, in 3 (4.2%) men with UA≤380 μmol/l and in 9 women (7.8%) with UA≤387 μmol/l. Accordingly, elevated levels of UA or HU were found in female patients only, with an average age of over 50 years (63.5 ± 12.3), average BMI of 29 ± 2.9 kg/m² and SBP of 140 ± 12.6 mmHg. The influence of clinical parameters was assessed using regression analysis, which showed that at 95% confidence level, the regression coefficient of BMI is statistically significant, and

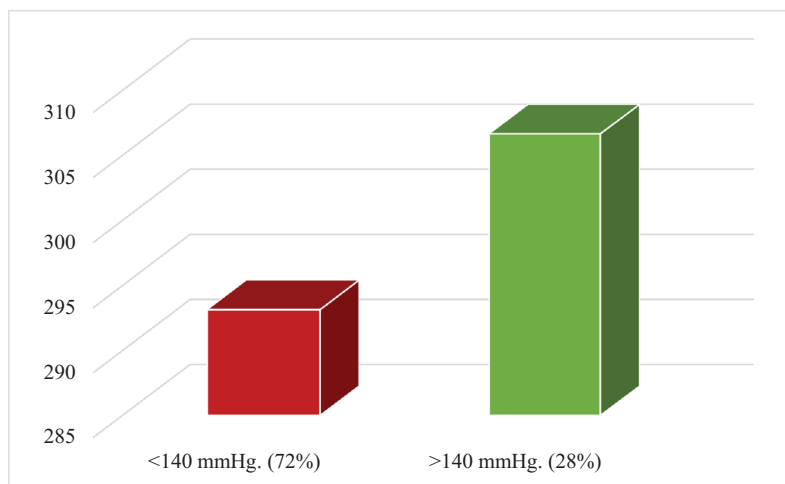


Figure 3. Distribution of participants by level of UA depending on SBP.

Source: compiled by the author.

Table 6. Regression coefficients of UA and SBP levels.

SBP	coefficient	p-value
<140 mmHg (n=135)	-0.14	0.25
>140 mmHg (n=52)	0.02	0.04

Source: compiled by the author.

with an increase in BMI by 1 kg/m², the level of UA increases by 1.83 µmol/l (p=0.006). The regression coefficients are presented in Table 7.

The study showed that with an increase in BMI by 1 kg/m², the level of UA significantly increases in patients with normal BMI, which may indicate potential metabolic disorders. However, the UA level had no statistically significant differences depending on the degree of hypertension. In patients with elevated SBP, a 1 µmol/L increase in UA was associated with a 2 mmHg increase in SBP.

The relationship between serum uric acid (UA) levels and body mass index (BMI) has been widely discussed in the literature, reflecting complex metabolic and physiological interactions. Uric acid is both a product of purine metabolism and a marker of oxidative stress, which can influence endothelial function, insulin resistance, and adipocyte activity (14,15). BMI, as a measure of overall body mass relative to height, is commonly used as a proxy for obesity and associated

Table 7. Regression coefficients on clinical characteristics of participants with HU.

	Coefficient	p-value
Age	0.21	0.36
BMI (kg/m ²)	1.83	0.006
SBP (mmHg)	0.01	0.91

Source: compiled by the author.

metabolic risk factors (16). Previous studies have suggested a positive association between elevated UA and higher BMI, implying that hyperuricemia may coexist with overweight and obesity due to increased purine turnover, decreased renal excretion, or pro-inflammatory mechanisms. However, these associations are not necessarily causal, as confounding factors such as age, sex, diet, comorbidities, and medication use can influence both UA and BMI levels, creating complex interdependencies that require careful statistical adjustment (17-20).

To address these complexities, the present analysis employed multivariable adjustment to account for potential confounding factors, including age, sex, blood pressure, and comorbid conditions. This approach allows for a more nuanced estimation of the independent relationship between UA and BMI while reducing bias from covariates that may otherwise distort observed

associations. The results indicated paradoxical patterns: certain subgroups demonstrated unexpectedly low or inverse correlations, despite an overall trend suggesting a positive association. Such paradoxes can arise when subgroup sizes are small or unevenly distributed, leading to statistical instability and reduced power to detect meaningful relationships. For instance, extreme BMI categories or age groups with limited representation can disproportionately influence regression estimates, amplifying apparent anomalies.

These findings underscore the importance of interpreting UA–BMI associations within the context of study design and sample characteristics. Small and potentially unbalanced subgroups limit generalizability and may exaggerate or obscure true effects. Furthermore, the presence of paradoxical results emphasizes that observational associations should not be over-interpreted as evidence of causation. Instead, these outcomes highlight the need for careful subgroup analysis, replication in larger cohorts, and complementary mechanistic research to clarify the physiological pathways linking uric acid metabolism with body composition. Overall, multivariable adjustment provides valuable insights but also reveals the methodological challenges inherent in analyzing interrelated metabolic markers in heterogeneous populations.

Discussion

The relationship between serum UA levels and hypertension remains understudied and ambiguous. It is believed that the prevalence of HU is higher in males than in females (21,22). The results of this study showed the opposite, that women have higher levels of UA than men (295 and 276 $\mu\text{mol/l}$, respectively). Most authors attribute this phenomenon to the effect of oestrogen on UA in women of reproductive age, and in the author's study, the average age of female patients was 57.5 ± 4.1 , hence a natural increase in the UA level with age. Other studies show that each 1 mg/dl increase in serum UA levels contributes to a 20% increase in the prevalence of hypertension in the general population without treatment for HU and AH (23–25). Moreover, studies show that the regression coefficients for UA and AH levels range from 14% to 20%. At the

same time, other studies indicate a negative correlation between hypertension and hypertension (26, 27). This diversity in results has been attributed by some authors to the same risk factors for increased UA and AH (13, 28, 29): overweight, excessive salt intake, diabetes mellitus, and vascular atherosclerosis.

This study showed no effect of overweight on serum UA levels. A trend towards higher UA was found in participants with normal body weight (BMI 21–25 kg/m^2) – the average UA value was 319 $\mu\text{mol/L}$. The average SBP level in this group was 128 mmHg with a maximum value of 150 mmHg. Several studies suggest that the presence of HU at normal body weight may predict the development of resistant AH (30,31). As such, Shi et al. (32) and Tian et al. (33) demonstrated the development of persistent AH in patients under 50 years of age and in adolescents with normal BMI, with increased UA concentrations.

The study yielded no definitive conclusion regarding the association between UA levels and increased SBP. Although among the gender, age and clinical characteristics of the patient studied, the association of UA level and SBP was seen in the group with persistently high blood pressure at the time of the study (>140 mmHg). Some studies also do not show this association or have low values for it. Moreover, Borghi et al. (34) observed that age-dependent UA levels are J-shaped, that is, they increase with patients' age. Similar results were obtained in the present study, where the level of HU was most pronounced in age groups over 50. The study found a tendency for UA levels in blood serum to increase with patient age (from 286 to 302 $\mu\text{mol/l}$), but statistical analysis did not confirm the significance of these changes. Although the regression coefficients of the HU dependence on age were not statistically significant ($p>0.05$), the trend towards the J-shape was evident. Nevertheless, the J-shape as a function of age in the study did not confirm the association of secondary HU with AH, and as many authors have argued, the rise in UA levels is related to age-related increase in cellular decay, and as a consequence of increased UA production (35).

The increased amount of UA in the blood serum affects the increase in SBP and diastolic BP, which was also mentioned by Bhosale et al. (36). The authors noted that the relationship between elevated blood

pressure and UA level can lead to hypertensive crises and pressure increases of more than 140 mmHg. However, the authors described 235 patients presenting with signs of hypertension, whereas the present study describes 187 patients and 1112 records on this problem. The authors' opinion that average systolic (153.2 vs. 149.8 mmHg) and diastolic (99.8 vs. 96.9 mmHg) BP was significantly higher in hyperuricaemia cases ($p < 0.01$) is noteworthy.

High levels of UA can identify the risk of developing hypertension in patients in time and enable appropriate preventive measures. This was also reported by Bhargava et al. (37). The authors noted that there is a direct link between UA levels and high blood pressure, but the study found that only 20% of the subjects had elevated UA levels, which determined that this is not always a direct sign of disease, which is noteworthy. Notably, elevated UA levels can be associated with metabolic syndrome, which also leads to complications in the form of hypertension. Therefore, all concomitant diseases should be addressed to identify a clear cause of a person's condition.

UA causes acute vasoconstriction and may be one of the actors in the increase in systolic and diastolic blood pressure, as Feig (38) also noted. The author also notes that since UA can affect the muscles of blood vessels and act quickly, elevated blood levels are more often detected in acute hypertension, and this phenomenon is rarely observed in hospitalised hypertension. With a stable increase in blood pressure, the level depends on sodium more than on UA, which is noteworthy. However, it is necessary to differentiate between hypertension as an independent disease in which the level of UA is important and elevated BP in other pathologies, for which an increase in UA levels is also characteristic. A qualitative and correct analysis of the anamnestic data and clinical parameters directly affects the effectiveness of treatment.

Increased UA levels in concomitant chronic or acute pathologies or pregnancy may indicate metabolic disturbances and lead to the worsening of the patient's condition, which was also noted by Singh et al. (39). The authors pointed out the importance of the indicator and its impact on women with pregnancy hypertension because it can lead to impaired foetal development, this statement is noteworthy. However, it

should be noted that the UA level should serve as an auxiliary laboratory test and should be based on associated symptoms, i.e., it is not the definitive test for the diagnosis of hypertension. These studies provide a balanced perspective on uric acid-lowering therapy in adults by evaluating neutral or negative outcomes in randomized controlled trials. Mackenzie et al. (40) investigated the effects of allopurinol on cardiovascular outcomes in patients with ischemic heart disease through the ALL-HEART RCT and found no significant reduction in major adverse cardiovascular events, indicating that lowering serum uric acid did not translate into clear cardiovascular benefit in this population. Similarly, Xu et al. (41) conducted a systematic review and meta-analysis of randomized trials assessing uric acid-lowering agents in patients with heart failure and reported mixed results, with many studies showing no significant improvement in clinical endpoints, highlighting the limited efficacy of these interventions in modifying cardiovascular outcomes. Together, these findings underscore that while uric acid-lowering therapy can reduce serum urate levels, its impact on major cardiovascular events remains uncertain, emphasizing the need to interpret associations between uric acid and hypertension or heart disease cautiously and within the context of broader evidence.

Notably, high UA levels can reduce vascular elasticity, causing the heart to work harder and increasing the risk of atherosclerosis and, as a result, hypertension. Moreover, high levels of UA can lead to sodium retention in the body, which can also increase blood pressure. Therefore, it is important to analyse the patient's condition, as comorbidities such as atherosclerosis, obesity, or diabetes can directly affect UA levels. This is why it is necessary to conduct a comprehensive examination of the patient to make a correct diagnosis and provide quality treatment.

Conclusions

Thus, not all patients with arterial hypertension (AH) in city polyclinics Nos. 1, 6, 7, and 10 who were part of the initial cohort were included in the analysis of serum uric acid (UA). True hyperuricemia (UA $> 360 \mu\text{mol/L}$ in women and $> 420 \mu\text{mol/L}$ in men) was

observed in a small subset of participants, specifically 9 women (7.8%). Regression analysis suggested a significant association between UA and BMI in women with hyperuricemia ($p=0.006$), and in patients with sustained elevated systolic blood pressure (>140 mmHg), a 1 $\mu\text{mol/L}$ increase in UA was associated with a 2 mmHg increase in SBP ($p=0.04$). These findings indicate potential relationships between UA, BMI, and blood pressure, but causal inferences cannot be drawn due to the study's cross-sectional design.

Important limitations include the lack of information on key clinical and lifestyle factors that may influence UA levels, such as renal function (eGFR or serum creatinine), use of diuretics or urate-lowering therapy, adherence to antihypertensive treatment, and diet or physical activity. Additionally, the absence of longitudinal follow-up prevents evaluation of temporal changes in UA or the impact of comorbid chronic conditions on serum levels. Therefore, the findings should be interpreted primarily as hypothesis-generating. Future research should focus on prospective, adequately powered cohort studies with systematic measurement of UA and comprehensive collection of clinical, therapeutic, and lifestyle data. Such studies are needed to clarify the potential prognostic value of UA and to inform evidence-based management recommendations, rather than to directly guide therapeutic or prognostic decisions at this stage.

Ethic approval: All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Conflict of interest: The author declares that she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' contribution: SA contributed to the study concept and design, analysed and interpreted the data, developed the theoretical framework and wrote the manuscript.

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